

Intervals and absolute value equations

Polynomial-size vertex test for inverse M-matrix intervals

Difficulty: challenging **Importance:** interesting to specialist

Status: [Lean verified](#)

Rating rationale: Replacing an exponential interval test by a quadratic vertex family is challenging; verified inverse-positivity has specialist importance in interval matrix analysis.

Area: interval linear algebra; structured matrices

Independently reviewed resolution - 2026-09-11

Affirmative resolution. Theorem 1 proves that every interval member is inverse-M if and only if the n^2 vertices $C - D_i R D_j$ are inverse-M. These are contained in the displayed two-sign family, so the original $2n^2$ equivalence follows. The proof covers all real endpoints, every dimension, zero widths, zero entries and reducible matrices without assuming regularity.

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. See the [complete manuscript](#), [independent agent review](#) and [submission record](#). The source archive identifies the drafts as AI-generated; authorship is recorded at the submitter's request. That original review was an informal agent review. The later Lean verification is recorded separately below; external human peer review is not claimed.

The difficulty, importance and rating rationale below are historical assessments of the original open target. The original statement and dated audits are preserved.

Lean proof and verification evidence

Mathematical argument: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Formalization:** Sidney Holden, with OpenAI Codex assistance, under Apache-2.0. **Integration and verification submission:** George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology. The [complete formalization at immutable revision cef3e2f4](#) imports Holden’s existing proof without changing its fourteen active Lean files.

The formal proof establishes the full original equivalence for **every positive dimension and every real closed entrywise interval**. In fact, the n^2 negative-sign vertices alone suffice, implying the original $2n^2$ two-sign criterion. Genuine invertibility and the signs of the actual inverse are included in the definition. Independent entries, zero widths, zero entries, reducible matrices, nonsymmetric endpoints and repeated vertices are all covered. No regularity or nonsingularity of the interval is assumed.

All **four declarations** are listed in [formalization.yaml](#):

- `NLA.IV03.vertex_formula`: the pointwise vertices equal the stated matrix formula.
- `NLA.IV03.vertices_admissible`: every specified vertex belongs to the interval.
- `NLA.IV03.nSquaredCriterion`: the stronger negative-sign vertex equivalence.
- `NLA.IV03.twoSignCriterion`: the complete original two-sign equivalence.

The [independent Challenge](#), [definitions](#), [frozen numerical targets](#) and [proof notes](#) record the mathematical boundary. Exact maximum principles, principal and Schur closure, complementary minors, adjugate completion and a resolvent identity avoid numerical interval grids. The Solution closure excludes Challenge’s deliberate specification placeholders. Complexity estimates are outside the formalized equivalence.

The project pins **Lean 4.33.1**, Mathlib `0df444a360eaa60ab8c11dca51a86af692955474` and LeanCert `621a43d7cf21f87872392a01e874f2f1dbddc926`.

Literal checked commit: `cef3e2f486285d0f6885231cda3ac2ff04c975cb`.

On **16 September 2026 UTC**, [Linux run 35059255598](#), [verification job 104675949451](#) checked that proof/integration revision. All four LeanCert kernel-trust assertions, Comparator statement checks and Lean default-kernel replay passed. Every exported theorem uses only `propext`, `Classical.choice` and `Quot.sound`. Required sandbox, rejection and checker controls ran inside the successful project job. [Actual logs and source hashes](#) bind all **140** candidate inputs.

Two independent AI-agent [complete-source reviews](#), an [import review](#), and an [audit of the fresh actual run](#) support this record. These are distinct from external human peer review or a guarantee that GitHub/checker software is infallible. No local macOS Lean execution is claimed. The [project README](#) gives `lake build Solution` and the shared Linux reproduction commands. Later publication and merge revisions require separate exact-commit checks; the named proof run does not certify a later revision.

The [bounded public duplicate audit](#) identified Holden’s existing formalization, which is deliberately reused and credited here. The original mathematical manuscript, canonical target, source reviews, frozen inputs and license remain unchanged.

Problem statement

A real square matrix M is a nonsingular M-matrix if its off-diagonal entries are nonpositive, M is invertible, and M^{-1} is entrywise nonnegative. An inverse M-matrix is the inverse of such a matrix.

For every $n \geq 1$, let $L, U \in \mathbb{R}^{n \times n}$ satisfy $L \leq U$ entrywise, and set

$$\mathcal{A} = \{A : L \leq A \leq U\}, \quad C = (L + U)/2, \quad R = (U - L)/2.$$

For $i = 1, \dots, n$, let $z^{(i)}$ have entry -1 in position i and entry 1 elsewhere, and write $D_i = \text{diag}(z^{(i)})$.

Is the following equivalence true?

$$\begin{aligned} & [\text{every } A \in \mathcal{A} \text{ is an inverse M-matrix}] \\ & \iff \\ & \left[\begin{array}{l} C + sD_iRD_j \text{ is an inverse M-matrix} \\ \text{for every } i, j \in \{1, \dots, n\} \text{ and } s \in \{-1, 1\} \end{array} \right]. \end{aligned}$$

All entries in the interval vary independently, and degenerate intervals are allowed. The proposed test uses at most $2n^2$ matrices; repeated test matrices are harmless. This would give a small exact certificate for a structured property needed in interval inversion and verified linear-system computation.

References

Milan Hladík, *An overview of polynomially computable characteristics of special interval matrices*, in *Beyond Traditional Probabilistic Data Processing Techniques*, Springer (2020), pp. 295–310; [author preprint](#), [arXiv:1711.08732v1](#), §9, Conjecture 1, p. 11, with the vectors defined in Theorem 24.

Jürgen Garloff, Doaa Al-Saafin, and Mohammad Adm, *Further Matrix Classes Possessing the Interval Property*, *Reliable Computing* **28** (2021), 56–70, p. 64, paragraph immediately before IP 4.4.

Status check

The 2021 paper explicitly retains this conjecture while proving a different sufficient vertex family of size 2^{n-1} . On 2026-09-10, searches combining the original title, inverse M-matrices, interval property, conjecture, and 2025/2026 found no resolution. Hladík’s current publication list and the later interval-property paper were checked. This is a selected-source status check, not an exhaustive citation audit.

Independent audit — 2026-09-10

The full Hladík preprint, §9, Conjecture 1, agrees with the displayed two-sign vertex family, including the definition of each $z^{(i)}$. Garloff–Al-Saafin–Adm, p. 64, expressly leaves that conjecture unresolved; its exponential family is a different test, not a resolution for a subclass here. Independent later searches for the exact conjecture and matrix class found no resolution.